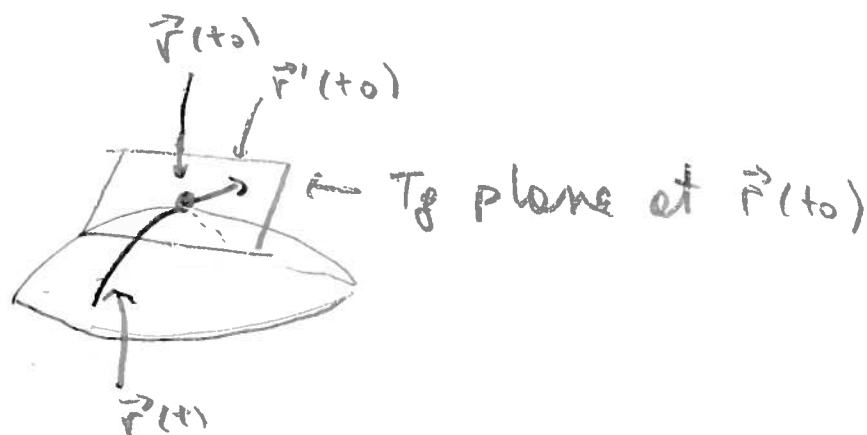


§12.7

We have already seen that the tangent plane to the surface $F(x, y, z) = \kappa$ at a point (x_0, y_0, z_0) is given by

$$\nabla F(x_0, y_0, z_0) \cdot (x - x_0, y - y_0, z - z_0) = 0$$

Thm If $\vec{r}(t)$ is a ^{diffe} curve contained in the ^{diffe} surface $F(x, y, z) = \kappa$, then $\vec{r}'(t_0)$ is contained in the tangent plane at $(x_0, y_0, z_0) = \vec{r}(t_0)$



Proof $F(\vec{r}(t)) = \kappa$ $\frac{\partial F}{\partial t} = \frac{\partial F}{\partial x} \frac{dx}{dt} + \frac{\partial F}{\partial y} \frac{dy}{dt} + \frac{\partial F}{\partial z} \frac{dz}{dt}$

where $\vec{r}(t) = (x(t), y(t), z(t))$ $\vec{r}'(t) = \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right)$

so $\frac{\partial F}{\partial t} = \nabla F \cdot \vec{r}'(t) \quad \square$

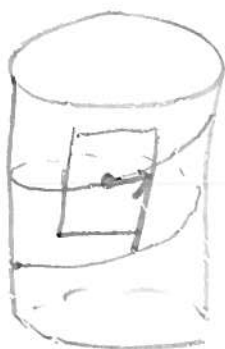
Ex $r(t) = (\cos t, \sin t, t)$ $F(x, y, z) = x^2 + y^2 = 1$

$t_0 = \frac{5\pi}{2}$ $\vec{r}(t_0) = (0, 1, \frac{5\pi}{2})$ $\nabla F = (2x, 2y, 0)$

$\vec{r}'(t_0) = (-\sin t_0, \cos t_0, 1) = (-1, 0, 1)$

$\nabla F(\vec{r}(t_0)) = (0, 2, 0)$

$\nabla F(\vec{r}(t_0)) \cdot \vec{r}'(t_0) = (0, 2, 0) \cdot (-1, 0, 1) = 0$



$F(x, y, z) = 1$

Differentials

$z = f(x, y)$

$dz = f_x dx + f_y dy = Df \cdot \langle dx, dy \rangle$

Reason $z - z_0 = f_x(x - x_0) + f_y(y - y_0) + \epsilon$

$\Delta z = f_x \Delta x + f_y \Delta y + \text{infinitesimal}$

Ex $z = 2x^3 + xy - y^3$ compute Δz & dz as (x, y) changes from $(2, 1)$ to $(2.03, 0.98)$

$\Delta z = f(2.03, 0.98) - f(2, 1) = 2 \cdot (2.03)^3 + 2.03 \cdot 0.98 - (0.98)^3 - (2^3 + 2 - 1)$
 $= 0.779062$

$dz = f_x dx + f_y dy = (6x^2 + y) dx + (x - 3y^2) dy = 25 dx - dy$

$$dx = 0.03 \quad dy = -0.02$$

$$dz = 25dx - dy = .77$$

Ex $P = T/V \quad T = 200 \text{ K} \quad V = 1 \text{ m}^3$

error on $T = \pm 1 \text{ K}$

on $V = \pm 0.05 \text{ m}^3$

$$dP = ?$$

Error on $P = ?$

$$dP \approx \frac{\partial (T/V)}{\partial T} dT + \frac{\partial (T/V)}{\partial V} dV = dT - \frac{T}{V^2} dV$$

$$\Delta P \approx \Delta T - \frac{T}{V^2} \Delta V = \pm 1 \mp \frac{200}{1} \cdot 0.05 = \pm 11$$

Ex 1) Find the eqn of tg plane at given pt.

a) $x^2 + y^2 + z^2 = 16 \quad (2, 3, \sqrt{3})$ b) $z = \frac{x^2}{4} + \frac{y^2}{4} \quad (2, 2, 2)$

c) $z = ze^{3y} \cos 2x \quad (\pi/3, 0, -1)$

Ex 2 Find dz and then Δz as (x, y) goes from P to Q

a) $z = 2x^2y^3 \quad P = (1, 1) \quad Q = (.99, 1.02)$

b) $z = \ln(x^2y) \quad P = (-2, 4) \quad Q = (-1.98, 3.96)$

c) $z = \tan^{-1} xy \quad P = (-2, -0.5) \quad Q = (-2.03, -0.51)$

Ex 3 The resistance is $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$ $I + R_1 = 25 \pm .5 \text{ ohm}$

What is R ?

$$R_2 = 100 \pm .5 \text{ ohm}$$